

MIMIC-III ICU Mortality Prediction Project

Pre-modeling data engineering, feature engineering
and data quality control

FIRST 24 HOURS

45,253 ICU STAYS

330M+ RECORDS

Goal: build a reliable machine learning dataset for in-hospital mortality prediction using the first 24 hours of clinical data after ICU admission.

1. Project goal and study cohort

Define the patient population and observation window before modeling.

45,253

Adult ICU stays

≥24 hours

Minimum ICU stay

First 24 hours

Feature time window

11.84%

In-hospital mortality

Cohort logic

- Include adult ICU stays lasting at least 24 hours, rather than all MIMIC-III records.
- Freeze the main cohort as the official ICU-stay list in a separate Parquet file.
- Restrict every feature table to these 45,253 ICU stays.
- Use the first 24 hours after ICU admission to reduce the use of information close to or after the outcome.

Why a fixed cohort?

- Distinguish missing measurements from genuine exclusions.
- Keep all sources within the same ICU population.
- Prevent lost stays during the final merge.

2. Identifiers and repeated ICU stays

A patient, a hospital admission, and an ICU stay are different units.

SUBJECT_ID

Patient identifier

HADM_ID

Hospital admission

ICUSTAY_ID

ICU stay

2,399

Admissions with multiple ICU stays

CHALLENGE

Joining by HADM_ID alone could assign measurements to the wrong ICU stay within the same hospital admission.

SOLUTION

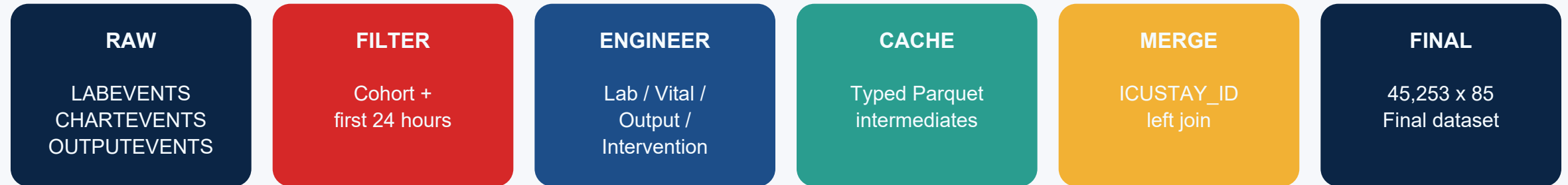
Add a time filter: link a measurement only to the ICU stay whose first 24-hour window contains it.

SUBJECT_ID > HADM_ID > ICUSTAY_ID + INTIME

Result: hospital-admission data is mapped to the correct ICU stay using identifiers and timing.

3. A modular clinical data pipeline

Turn raw MIMIC tables into independent, reusable feature tables.



CHALLENGE

Reprocessing very large CSV files in every notebook was slow and put pressure on memory.

SOLUTION

Save typed Parquet outputs after each major stage; later stages read the cleaned tables directly.

Saved feature tables

lab_features_first24h.parquet
vital_features_first24h.parquet
output_features_first24h.parquet
intervention_features_first24h.parquet
demographic_features.parquet

Rebuild the final dataset without scanning the raw tables again.

4. Laboratory feature selection

Build a clinically meaningful core set from hundreds of tests.

726

Distinct laboratory ITEMIDs

~60

After the 50% coverage filter

23

Clinically selected tests

19

Final laboratory tests

Selection logic

- Measure coverage by distinct hospital admissions (HADM_ID), not by row count.
- Review common tests for clinical relevance and redundancy.
- Simplify overlapping signals: hemoglobin/hematocrit/RBC, INR/PT/PTT, and glucose variants.
- Remove AST, ALT, bilirubin, and albumin due to high missingness; retain lactate and blood gases for clinical relevance.

Final core tests

- Hb, WBC, platelets, creatinine, BUN, sodium, potassium, chloride, bicarbonate, anion gap, glucose, magnesium, calcium, phosphate, INR, pH, pO₂, pCO₂, and lactate.

5. Laboratory data engineering

Link nearly 10 million selected records to the correct ICU stay and time window.

9,978,641

Selected laboratory records

1,735,994

First 24-hour records

500K

Rows per processing chunk

44,626 x 39

Final laboratory table

CHALLENGE

Merging about 10 million rows with the ICU map in one operation caused WSL/RAM crashes.

SOLUTION

Process LABEVENTS in 500K-row chunks, match HADM_ID, and apply the CHARTTIME / INTIME + 24-hour filter.

Feature design

first

Initial measurement

last - first

Change during the window

**Negative changes are not errors.
They describe clinical direction.**

6. Vital signs: processing 330M+ records

Memory-friendly streaming for event data beyond a single in-memory read.

330,712,483

CHARTEVENTS source rows

7,031,935

Selected first-day vital events

44,549 x 33

Final vital-sign table

8

Canonical vital-sign groups

CHALLENGE

The source presentation reports parsing/memory failures after 5-6 minutes with 500K-row pandas chunks.

SOLUTION

Scan in small blocks using strict PyArrow streaming, retain selected ITEMIDs, and create a Parquet cache.

CareVue + MetaVision

- Map equivalent ITEMIDs across systems to one canonical feature.
- Use heart rate, respiratory rate, SpO2, non-invasive SBP/DBP/MBP, temperature, and GCS total.
- Normalize Celsius/Fahrenheit units before aggregation.
- Summarize each vital sign with first, last, minimum, and maximum values.

7. Output features: urine, chest tube, and EBL

Preserve the meaning of different output-event recording patterns.

Urine output

- Combine Foley, voided, operating-room, and other relevant urine ITEMIDs.
- Sum the first 24 hours per ICU stay; preserve 2,750 genuinely missing values.
- Remove a -1,500 mL record: the stay total changes from -300 to 1,200 mL.
- Remove a 555,975 mL gross error: 561,190 becomes 5,215 mL.

Chest tube

- Amount data is missing for more than 82% of the cohort.
- Use a presence/absence indicator instead of imputing an unreliable amount.
- 8,106 ICU stays have chest_tube_present = 1.
- Record presence was considered more reliable than sparse volume data.

Estimated blood loss

- Keep ebl_present and ebl_24h together; 4,291 stays have EBL records.
- Median about 300 mL; maximum 50,000 mL.
- p99: 10,000; p99.5: 13,685; p99.9: 25,000 mL.
- If ebl_present = 0, set ebl_24h = 0.
- Retain extremes and flag them for modeling review.

8. Intervention and demographic features

Add clinically informative variables without expanding the project scope.

Intervention features

47.86%

Mechanical ventilation

30.20%

Vasopressor use

2.57%

RRT / CRRT

Sources: CHARTEVENTS, INPUTEVENTS_CV/MV, PROCEDUREEVENTS_MV, and OUTPUTEVENTS. Combine CareVue and MetaVision evidence into one binary feature per intervention.

Demographics + admission

- age, gender, admission_type, admission_location

Intentionally excluded

- Insurance, marital status, ethnicity, religion, and language

Age policy

- INTIME - DOB; cap ages 90+ at 90 due to deidentification (2,239 stays).

9. Missingness: preserve its meaning

Missing measurements can carry clinical signal in ICU data.

Lab / vital coverage	Stays	Mortality
Both present	44,187	11.8%
Lab only	439	12.3%
Vital only	362	22.7%
Both missing	265	1.9%

Decision

- Do not remove stays because of missingness.
- Add group-level lab_missing and vital_missing indicators.
- Preserve individual lab/vital NaNs; defer median imputation to modeling.
- Handle EBL separately: ebl_present = 0 implies ebl_24h = 0.

Why it matters: 22.7% mortality in the lab-missing / vital-present group suggests that missingness is not completely random.

10. Data quality: outliers versus errors

Fix issues upstream, at the source stage, before rebuilding feature tables.

CHALLENGE

Invalid vital values included negative Celsius temperatures, HR 5,409, RR 2,355,555, SpO2 981,023, and BP above 120,000.

SOLUTION

Apply native-unit limits before conversion/aggregation; set ERROR-marked values to NaN. Update 6,840 vital feature cells.

CHALLENGE

Laboratory records included clinically invalid zero hemoglobin and zero INR.

SOLUTION

Invalidate only clearly impossible Hb/INR values at or below zero. Update six feature cells.

CHALLENGE

Urine records contained gross errors of -1,500 mL and 555,975 mL.

SOLUTION

Filter negative and gross-error records upstream; regenerate the output-feature Parquet file.

Intentionally retained: negative changes, negative anion gap, plausible extreme lab/vital values, EBL up to 50,000 mL, and some very high urine totals.

11. Leakage prevention and final validation

Prevent the model from using future outcome information.

Outcome information excluded

- DOD, DOD_HOSP, and DOD_SSN
- DEATHTIME
- DISCHTIME and DISCHARGE_LOCATION
- EXPIRE_FLAG-type outcome fields
- Use HOSPITAL_EXPIRE_FLAG only as the target.

Final validation

- 45,253 unique ICUSTAY_ID values
- Zero duplicate ICU stays, missing targets, or infinite values
- No suspicious outcome-related predictor columns beyond the target
- Regenerate source tables after source-level cleaning.

12. Final dataset and transition to modeling

Data engineering and feature engineering are complete.

45,253

Rows / ICU stays

85

Total columns

83

Predictors

5,360

Mortality-positive stays

Final data structure

- Lab, vital, output, intervention, and demographic features in one table
- Left joins on ICUSTAY_ID
- Retain lab_missing and vital_missing
- Handle EBL missingness separately
- Preserve 424,942 clinical missing cells

Next step: modeling

- Median imputation fitted during preprocessing
- Categorical encoding
- Patient-level train/validation/test splits
- Compare multiple models
- Evaluate class imbalance and outlier strategies

What does this project demonstrate?

LARGE-SCALE DATA

Memory-friendly streaming and chunk processing across 330M+ records.

CLINICAL FEATURE ENGINEERING

Select meaningful tests and engineer time-window and trajectory features.

DATA QUALITY

Resolve gross errors, unit issues, missingness, and outlier distinctions at source.

LEAKAGE PREVENTION

Exclude post-outcome information and preserve the first 24-hour window.

MODULAR PIPELINE

Build a reusable, auditable workflow with Parquet intermediate tables.

PROBLEM SOLVING

Address memory failures, parsing errors, repeated stays, and inconsistent data.

Result: a reliable, traceable, and clinically meaningful dataset, ready for modeling.